

MintpathOS

The First Biologic Autonomous Kernel

December 31, 2025

Production Validated (Revenue Generating)

Patent Pending

Classification: Level 3 Autonomous Infrastructure

This document describes system behavior and architecture. Certain mechanisms are intentionally abstracted.

1. Executive Summary

MintpathOS is the first production-validated Level 3 Autonomous Kernel capable of planning, executing, evaluating, and safely evolving its own production codebase under strict governance, legality, and audit constraints. Level 3 autonomy denotes systems that independently operate and adapt without human-in-the-loop approval for routine evolution, while remaining bounded by non-overridable policy constraints.

Built by a single operator without formal software training in approximately five weeks, MintpathOS demonstrates that governed autonomous systems can compress traditional engineering timelines by orders of magnitude. Total development cost was minimal: a \$5 Railway hobby subscription (used briefly) and a few hundred dollars in API usage.

MintpathOS is live in production, generates revenue, and includes both CLI and mobile PWA interfaces for governed operational control.

2. System Architecture Overview

MintpathOS is structured as a multi-layered autonomous organism designed to maintain coherence, safety, and deterministic evolution.

2.1 SQL Skeleton (Structural Truth)

The SQL layer forms the immutable backbone of system reality through an Auto-Safe SQL Evolution Engine enforcing additive-only evolution, atomic migrations, rollback isolation, and hard invariants.

2.2 Semantic Control Plane (Cognition)

The TypeScript control plane acts as a bilingual reasoning layer bridging rigid SQL structure with dynamic logical meaning, coordinating job pipelines and worker nodes.

Cerebrum is the long-lived authoritative reality substrate, exposing queryable faculties (RepoFaculty, DbFaculty, FsFaculty) to ensure fresh, tool-backed reads without serialized snapshots or drift.

2.3 Liquid Crystal Architecture (Schema Immune System)

Liquid Crystal enforces coherence across SQL schemas, TypeScript types, JSON contracts, and runtime validators to prevent drift and silent failures.

2.4 Architect Router (Self-Healing Dispatch)

The Architect Router autonomously detects malformed or stalled intentions and normalizes them into executable envelopes.

Padlock Substrate provides a drift-resistant execution layer with hash-locked JobSpecs, restricted IR opcodes, mandatory verifiers, and fail-closed invariants.

PatchPlan defines the canonical mutation primitive: idempotent, multi-file plans with runtime validation and quarantine-backed execution.

3. Recursive Coherence

MintpathOS exhibits Recursive Coherence—the capacity to recognize and repair its own schemas, pipelines, and reward mappings. This is demonstrated in production via full self-evolution cycles: intent → governed patch → root build → auto-commit/push.

4. Governor: Cryptographic Execution Authority

Governor is the sole authority for state mutation, enforcing signed envelopes, deterministic payloads, and regulator-grade auditability with immutable logs.

5. M-CON v2: Intent to Operation Compiler

M-CON v2 replaces free-form prompting with deterministic compilation of intent into bounded, typed operations, allowing natural language job execution without brittle JSON prompts.

6. LUX: Governed UI Transform Kernel

LUX governs visual transformations via registered ops, visual anchors, and risk-based dampening. The MintOps PWA provides secure mobile access to governed operations.

7. Biologic Affective Control Substrate

MintpathOS maintains a deterministic affective state modulating execution posture using arousal, valence, trust, and workload metrics grounded in first-party datasets.

8. Grounded Model Pipeline

All model inference and learning rely on first-party logs, deterministic snapshots, and governed lineage to structurally bound hallucinations.

9. Conclusion

MintpathOS establishes a new paradigm for autonomous infrastructure: biologic control with structural rigor, preserving human sovereignty while enabling safe autonomous evolution.

Appendix A: Deterministic Replayability

Every autonomous action is replayable via cryptographic snapshots, pure kernel functions, and immutable ledgers.